Manufacturing & Costing Architecture

Planning Families, Cost Versions, BOM Generation & Gap Analysis

Kebony — Odoo 19 | Module: `kebony_manufacturing` | March 2026

1. Manufacturing Hierarchy

The atomic planning, costing, and valuation unit is the **Dryer Load**. One Dryer Load = One Manufacturing Order.

PHYSICAL HIERARCHY

```
Raw Lot (white wood, received at FIFO)
  ↓
Process Pack (PP)     [physical only]
  ↓
Autoclave Batch (AC) [5–8 packs, 2–8h cycle]
  ↓
Dryer Load (DL)       [2 AC = 1 DL, 40–70h]
  ↓                        = 1 MO
Finished Lot          [sellable FG]
  ↓ (optional)
RDK Machining         [subcontracting MO]
```

ODOO MODELS

MODEL	ACCOUNTING?	STATUS
`kebony.planning.family`	No — master data	DONE
`kebony.process.pack`	No — physical only	DONE
`kebony.autoclave.batch`	No — physical only	DONE
`kebony.dryer.load`	**Yes** — = MO	DONE
`mrp.production`	Yes — WIP + FG	PHASE 6
`kebony.cost.version`	Yes — budget layer	DONE

> **Key rule:** Only products from the **same planning family** can share an autoclave load or dryer load.

2. Planning Families — The Template Layer

A planning family groups products by species + dimensions + impregnation/drying profile. It defines all production parameters used for BOM generation and costing.

WHAT A FAMILY STORES

PARAMETER	UNIT	USED FOR
Stacking hours ref	hours	BOM operation
Autoclave hours ref	hours	BOM operation
Dryer hours ref	hours	BOM operation
Capacity weight	dryer h/m³	DL scheduling
Typical dryer load volume	m³	Cost per m³ calculation
Mix consumption per m³	kg/m³	BOM chemical line
Scrap internal %	%	Scrap factor → BOM wood qty
Scrap B-grade %	%	Scrap factor → BOM wood qty
Raw wood product	product	BOM wood component
Ready-mix product	product	BOM chemical component

FAMILY TYPES

SCOTS PINE (KSP)

55–80h dryer, 1.34–1.96 h/m³
230–309 kg/m³ chemical
4 FAMILIES

RADIATA (KRRS)

48–54h dryer, 1.06–1.30 h/m³
541–677 kg/m³ chemical
7 FAMILIES

SPECIALTY (OTHER)

30–210h dryer, 0.77–14.74 h/m³
Wide range
5 FAMILIES

Scrap factor = 1 / (1 – internal% – b_grade%)
E.g. 4.7% internal + 5.9% B-grade → scrap factor = **1.118**
Meaning: 1.118 m³ raw wood needed per 1 m³ FG output.

3. BOM Generation — From Family to Product

The **"Push BoM to Products"** button on a planning family creates or updates the BOM for every product assigned to that family.

WHAT GETS GENERATED

BOM ELEMENT	SOURCE	QTY FORMULA
Wood component	`raw_wood_product_id`	`scrap_factor` per 1 unit FG
Chemical component	`ready_mix_product_id`	`mix_consumption_per_m3` (kg)
Stacking operation	WC: STACK	`stacking_hours_ref` × 60 min
Autoclave operation	WC: AC	`autoclave_hours_ref` × 60 min
Dryer operation	WC: DRY	`dryer_hours_ref` × 60 min

> **Maintenance rule:** Change parameters on the *family*, then re-push. Never edit individual product BOMs directly — they will be overwritten on next push.

EXAMPLE: KRRS 1" (RADIATA)

```
Planning Family: KRRS 1" (KRRS1)
Scrap: 2.3% internal + 1.6% B-grade
Scrap Factor: 1 / (1 - 0.039) = 1.0406

BOM for product "KRRS 50x150mm":

┌──────────────────────────────────────┐
│ Component: Raw Radiata   qty: 1.0406 m³ │
│ Component: Ready-Mix    qty: 676 kg    │
│ Op 1: Stacking          4.5h (270 min) │
│ Op 2: Autoclave         0h   (per AC)  │
│ Op 3: Dryer             54h  (3240 min)│
└──────────────────────────────────────┘
┌──────────────────────────────────────┐
│ Output: 1 m³ FG (KRRS 50x150mm)          │
└──────────────────────────────────────┘
```

4. Cost Version Framework — The Budget Layer

A **Cost Version** stores period-versioned budget and standard cost references. It is the "what should it cost" baseline for gap analysis.

HEADER: KEBONY.COST.VERSION

FIELD	DESCRIPTION
`version_type`	Budget / Standard / Forecast
`fiscal_year`	Year (e.g. 2026)
`date_from / date_to`	Monthly period boundaries
`locked`	Approved → read-only

4 CHILD LINE TYPES

CHILD	PURPOSE
BU BUDGET	Top-down: BU + GL account + € amount
WC COST LINE	Bottom-up: WC + GL account + €/hour
PRODUCT PRICE	Standard RM price per UoM
FAMILY CARD	Hours + scrap snapshot + computed costs

LIFECYCLE

```
1. Create version for a month (e.g. July 2026)

2. Populate
   "Snapshot Families" button → family lines
   Manually add WC lines + product prices

3. Review
   Computed cost card auto-updates
   Verify total standard cost per m³

4. Lock & Approve
   Version becomes read-only
   Now visible on product cost cards
   Used as baseline for gap analysis

5. Next period
   "Duplicate Version" → adjust prices/rates
   Lock the new one
```

5. The Closed Loop — Work Centre × GL Account

Each WC cost line links a **work centre** to a **GL account** (cost nature) with a rate per hour. This is the key to variance decomposition.

EXAMPLE: DRYER WORK CENTRE

GL ACCOUNT	COST NATURE	€/HOUR
620000	Personnel (operators)	25.00
610000	Energy (electricity)	12.50
630000	Depreciation (dryer asset)	8.00
611000	Maintenance	3.50
Total Dryer Rate		**49.00**

> **Why multiple lines per WC?** Each GL account is a separate cost nature. When the actual GL balance is compared to absorbed amount (rate × hours), you get **per-nature variance**: "Energy was over budget by €2k, but personnel was under."

VARIANCE DECOMPOSITION

```
Budget (from cost version):
  Dryer rate    = €49.00/h
  Dryer hours   = 54h (from family KRRS1)
  Load volume   = 49.4 m³
  Budget cost   = 49 × 54 / 49.4 = €53.56/m³

Actual (from MO completion):
  Actual hours  = 58h (4h over)
  Actual GL 620 = €1,600 (vs budget €1,350)
  Actual GL 610 = €700  (vs budget €675)

Variances:
  Efficiency: (58−54)h × €49/h = €196 unfav.
  Rate (620): €1,600 − (58 × €25) = €150 unfav.
  Rate (610): €700 − (58 × €12.5) = −€25 fav.
```

6. Family Cost Card — Standard Cost per m³

Each family line in a cost version automatically computes the full standard cost breakdown. This is the bridge between planning families and financial control.

COMPUTED FIELDS (PER M³ OF FG)

COST ELEMENT	FORMULA	EXAMPLE (KRRS 1")
Scrap factor	1 / (1 – int% – bg%)	1 / (1 – 0.016 – 0.023) = 1.0406
Wood cost / m³	scrap_factor × wood_price	1.0406 × €280 = €291.37
Chemical cost / m³	mix_kg_per_m3 × mix_price_kg	676 × €0.045 = €30.42
Stacking / m³	stack_h × Σ(stack rates) / vol	4.5 × €35 / 49.4 = €3.19
Autoclave / m³	ac_h × Σ(ac rates) / vol	0 × €42 / 49.4 = €0.00
Dryer / m³	dry_h × Σ(dry rates) / vol	54 × €49 / 49.4 = €53.56
Conversion / m³	stack + ac + dryer	€3.19 + €0.00 + €53.56 = €56.75
Total std cost / m³	**wood + chem + conv**	**€378.54**

DATA SOURCES

HOURS & SCRAP

From family line snapshot (editable override of family master)

WC RATES (€/HOUR)

Sum of all WC cost lines for each work centre (multi-line per GL account)

VOLUME (M³)

`typical_dryer_load_volume` from planning family master

RM PRICES

From Product Price lines in the same cost version (wood & ready-mix)

All computed, all stored. Change a WC rate or RM price → family cost card updates automatically. Lock the version → snapshot frozen.

7. Product Cost Card — Standard vs Actual

Every product with a planning family shows a **standard cost card** on its form, pulled from the active (locked) cost version. This enables instant comparison with FIFO actual cost.

PRODUCT FORM — PLANNING TAB

PLANNING FAMILY: KRRS 1"

Stacking hours	4.5h
Dryer hours	54h
Mix consumption	676 kg/m³
Scrap (int + bg)	1.6% + 2.3%

STANDARD COST CARD (FROM BUDGET H2-2026)

Std Wood Cost / m³	€291.37
Std Chemical Cost / m³	€30.42
Std Conversion Cost / m³	€56.75
Std Total Cost / m³	**€378.54**

GAP ANALYSIS LOGIC

DIMENSION	BUDGET (COST VERSION)	ACTUAL (FIFO MO)
Wood price	Product Price line	FIFO purchase price
Wood qty / m³	Scrap factor (family)	Actual MO consumption / output m³
Chemical price	Product Price line	FIFO ready-mix cost
Chemical qty / m³	Mix consumption (family)	Actual MO consumption / output m³
WC rate / hour	WC Cost Lines (per GL)	Actual GL balance / actual hours
WC hours	Family reference hours	Actual MO operation hours

Price variance = (actual price – std price) × actual qty
Volume variance = (actual qty – std qty) × std price
Efficiency variance = (actual hours – std hours) × std rate
All decomposable by work centre, by GL account, by family.

8. Data Architecture — Where Is What Stored?

DATA	MODEL	MAINTAINED BY	WHEN UPDATED
Production parameters (hours, scrap, chemical)	`kebony.planning.family`	Production manager	When process changes (rare)
Raw wood & chemical product links	`kebony.planning.family`	Production manager	Once per family (setup)
Product BOMs (wood, chemical, operations)	`mrp.bom`	**Auto-generated** (Push BoM button)	After any family parameter change
WC hourly rates (by GL cost nature)	`kebony.cost.version.wc.line`	Controller / CFO	Monthly (new cost version)
RM standard prices (wood, chemical)	`kebony.cost.version.product.line`	Controller / Purchasing	Monthly (new cost version)
BU budgets (top-down)	`kebony.cost.version.bu.line`	CFO / Finance	Monthly or quarterly
Family cost card (std cost per m³)	`kebony.cost.version.family.line`	**Auto-computed**	On any input change (stored)
Product cost card (std cost per m³)	`product.template` (computed)	**Auto-computed**	Real-time (non-stored)
Actual production costs	`mrp.production` + FIFO layers	System (MO completion)	At MO closing

Golden rule: Parameters live on the **family**. BOMs are **generated**, not manually maintained. Prices & rates live on the **cost version**. The cost card is **computed**. Never edit downstream — always change the source.

9. Maintenance Workflow — Who Does What, When

PRODUCTION MANAGER

- → Update family parameters when process changes
- → Set `raw_wood_product_id` per family (once)
- → Click **"Push BoM to Products"** after changes
- → Assign planning family to new products
- → Review scrap % quarterly with quality team

> **Frequency:** Rare — only when a production process actually changes (new recipe, equipment upgrade, different chemical mix).

CONTROLLER / CFO

- → Create new cost version each month
- → Duplicate previous month → adjust
- → Update WC rates if cost pool changes
- → Update RM prices from latest PO data
- → Review family cost card totals
- → **Lock & Approve** when satisfied

> **Frequency:** Monthly — part of EOM close process. Lock before running margin reports.

SYSTEM (AUTOMATIC)

- ✓ Family cost card: auto-computed on input change
- ✓ Product cost card: auto-computed from active version
- ✓ BOM wood qty: auto from scrap factor
- ✓ BOM operations: auto from family hours
- ✓ BOM chemical: auto from mix consumption
- ✓ FIFO layers: auto at goods receipt
- ✓ MO actual cost: auto at MO completion

10. Chemical Supply Chain — Separate Internal Flow

The chemical mix has its own internal supply chain. The autoclave never consumes raw chemicals directly — only ready-mix.

FLOW

```
Raw Chemicals (FIFO valued)
  Water:              17,680 liters
  Furfuryl Alcohol:    9,821 kg
  Maleic Anhydride:      250 kg
  Citric Acid:           147 kg
  Sodium Carbonate:      145 kg
        ↓
Mix Production MO
  Consumes raws → Produces Ready-Mix
  1 batch = 28,043 kg
        ↓
Ready-Mix Tank (stock location)
  Triggered by minimum tank level
        ↓
Autoclave Consumption
  BOM line: mix_consumption_per_m³ (kg)
```

COSTING IMPLICATION

READY-MIX VALUATION

FIFO — each Mix MO batch creates a FIFO layer at actual raw chemical cost + absorbed WC (Chemical Mixing work centre).

STANDARD CHEMICAL COST

Cost version stores a `ready_mix_price_per_kg` via the Product Price line. Family cost card computes: mix_consumption × price = chemical cost per m³.

VARIANCE

Actual ready-mix FIFO cost vs standard price = **chemical price variance**. Actual kg consumed per m³ vs standard = **chemical efficiency variance**.

11. The Full Picture

```
MASTER DATA (rarely changes)
  Planning Family    → hours, scrap %, chemical rate, raw_wood_product, ready_mix_product
  Work Centres       → Stacking, Autoclave, Dryer, Chemical Mixing (physical definitions)

BUDGET LAYER (monthly, by controller)
  Cost Version       → period + version_type + locked status
    ├ BU Budget Lines  → BU (analytic) + GL account + planned €
    ├ WC Cost Lines    → WC + GL account + €/hour        ← THE CLOSED LOOP
    ├ Product Prices   → product + standard price / UoM
    └ Family Cards     → hours + scrap snapshot + computed cost per m³

BOM LAYER (auto-generated from family)
  Push BoM to Products → wood (scrap-adjusted) + chemical + 3 WC operations
  One BOM per product, template from family

OPERATIONAL LAYER (production)
  Dryer Load → Autoclave Batches → Process Packs → MO (Phase 6)
  Actual hours, actual consumption, actual FIFO cost

GAP ANALYSIS
  Budget (cost version)  vs  Actual (MO + FIFO)
  → Per product, per family, per WC, per GL account
  → Price / Volume / Efficiency variance decomposition
```

Everything is connected. Change a WC rate → family cost card updates → product cost card updates. Change scrap % → BOM wood qty updates → cost card updates. Lock the version → snapshot frozen for gap analysis. FIFO provides the truth. The cost version provides the benchmark.